Scheduling long-haul battery-electric trucks: coordinating charging, breaks, and daily rests

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Abstract

Recent advances in battery technology and fast charging are making long-haul battery-electric freight transport increasingly feasible. As vehicle ranges extend beyond regional distribution, charging decisions become increasingly intertwined with Hours-of-Service regulations. Since charging is time-consuming and unavoidable on long routes, operators must ensure that vehicles remain sufficiently charged while complying with driver regulations. Efficient operation therefore depends on synchronizing charging activities with mandatory driver breaks and daily rest periods. We study this scheduling problem for battery-electric trucks operating on multi-day long-haul routes. We introduce a multi-day electric truck driver scheduling problem in which charging decisions are optimized jointly with Hours-of-Service compliance to minimize total route duration. The problem is formulated as a resource-constrained shortest path problem and solved using an exact labeling algorithm that jointly models battery dynamics and driver regulations. Computational experiments show that optimized battery-electric truck schedules can, under favorable charging conditions, be completed within XX% of the duration of equivalent diesel-truck operations. However, schedule feasibility and performance deteriorate rapidly when charging opportunities are sparse or charging power is insufficient. Compared with a simpler charging strategy, the proposed optimization approach reduces route durations by up to XX%. Moreover, allowing partial recharging provides modest but consistent reductions in route time relative to full recharging. The results demonstrate that synchronization of charging and mandatory driver interruptions is a key operational enabler of long-haul freight electrification.

Keywords: Battery-electric trucks; long-haul freight transport; hours-of-Service regulations; charging scheduling; driver scheduling; resource-constrained shortest path.

1. Introduction

The electrification of road freight transport has progressed rapidly over the last decade. Battery-electric trucks are now being adopted in urban distribution and regional logistics, where vehicles can

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typically recharge at depots or terminals during predictable off-duty periods (Shoman et al., 2023). Recent improvements in battery capacity and fast-charging technology are extending feasible driving ranges and making battery-electric trucks increasingly viable for long-haul freight operations. Current commercial vehicle offerings illustrate this development: heavy-duty battery-electric trucks are commonly marketed with single-charge ranges of around 300 km, while newer long-haul models target approximately 500 km under favorable operating conditions (Volvo Trucks, 2025b; Mercedes-Benz Trucks, 2025). However, road freight transport is a highly competitive business with low profit margins (Council of the European Union, 2025), where schedules are tight and time losses translate directly into cost. If charging causes additional time losses, battery-electric trucks can quickly become unattractive for long-haul freight operations.

A defining feature of long-haul trucking is that operations are constrained by Hours-of-Service (HoS) regulations. Under EU regulation, drivers must take mandatory breaks and daily rest periods, typically taken overnight, that reset driving and working-time limits (Goel & Irnich, 2017). Traditionally, compliance with these regulations has been the primary scheduling challenge, as diesel refueling is fast and widely available. Consequently, refueling is rarely modeled as a binding scheduling resource in conventional truck operations (Schulz, 2024). For battery-electric trucks, in contrast, charging is substantially more time-consuming than diesel refueling and depends on the availability of suitable charging infrastructure. Operators must therefore ensure that vehicles remain sufficiently charged while simultaneously complying with driver regulations. Efficient operation depends on synchronizing charging activities with mandatory breaks and daily rests such that charging can be absorbed into periods during which the driver is already required to be off duty.

In this paper, we investigate the synchronization of charging activities with mandatory breaks and daily rests in long-haul battery-electric freight transport. We study a multi-day scheduling problem in which charging, breaks, and daily rests must be jointly planned to minimize route duration while satisfying both Hours-of-Service and energy constraints.

1.1. Related literature

The Truck Driver Scheduling Problem (TDSP) concerns the construction of driver schedules that satisfy regulatory HoS constraints while optimizing operational objectives such as route duration or transportation cost. The Truck Driver Scheduling Problem (TDSP) has been studied extensively under Hours-of-Service regulations (Archetti & Savelsbergh, 2009; Goel, 2012; Goel & Kok, 2012).

Methodologically, TDSPs are commonly formulated as shortest path problems with resource constraints (SPPRCs). Feasible schedules are constructed incrementally through labeling algorithms that extend partial paths while tracking resources such as accumulated driving and working time since the last break or rest. Resource extension functions are used to update resource states and identify infeasible path extensions (Goel, 2010; Drexl & Prescott-Gagnon, 2010). Building on this paradigm, Goel & Irnich (2017) introduced an exact branch-and-price framework with a parameter-free network representation that avoids time discretization, while preserving regulatory feasibility. However, these models assume conventional vehicles with negligible refueling times and do not consider energy constraints.

While TDSP research focuses on regulatory feasibility, the Electric Vehicle Routing Problem (EVRP) literature provides a rich body of work on modeling battery dynamics and charging decisions (Schneider et al., 2014; Han et al., 2025). Many exact EVRP approaches are based on branch-and-price frameworks in which a master problem selects vehicle routes, while SPPRC pricing problems generate new feasible routes. In these formulations, battery state is treated as a resource and charging duration directly affect route timing (Desaulniers et al., 2016; Lam et al., 2022; Nafstad et al., 2025). Several works allow partial recharging and nonlinear charging functions to better reflect operational practice (Montoya et al., 2017; Froger et al., 2019). Whereas most EVRP studies focus on route construction and charging decisions along a route, recent work has begun to integrate routing decisions with the scheduling of charging infrastructure. For example, Yamín et al. (2025) jointly optimize urban delivery routes and overnight depot-charging schedules using a branch-price-and-cut algorithm. Their formulation accounts for nonlinear charging and limited charging capacity, but does not consider HoS-constrained driver schedules.

Only a limited number of studies explicitly consider the interaction between charging activities and driver breaks and rests under HoS regulation. Schiffer et al. (2017) extend the EVRP by incorporating HoS regulations and allowing en-route charging activities to be synchronized with mandatory driver breaks. Their EVRP is motivated by one-day routing and scheduling of electric commercial delivery vehicles in regional distribution networks. They solve the problem using a mixed-integer formulation combined with an adaptive large neighborhood search heuristic. Closer to the long-haul setting considered in this paper, Wan et al. (2024) study an opportunity-charging planning problem for a battery-electric truck in a single freight corridor case study. Their model coordinates charging decisions with planned stops at highway service areas and logistics hubs while accounting for charging technologies, electricity prices, and temporal flexibility. However, HoS regulations are represented only through simplified driving-time and rest constraints and the model does not consider recurring daily-rest requirements. More recently, Brockmann & Guajardo (2025) develop a mathematical programming model for an Electric Vehicle Truck Driver Scheduling Problem (EVTDSP) in which charging operations and mandatory driver breaks are jointly optimized. Their analysis highlights the importance of accounting for driver regulations when assessing the operational performance of battery-electric trucks. However, the study is restricted to single-day operations and does not consider recurring daily rests or multi-day scheduling decisions. Finally, Schneider et al. (2023) analyze one-day electric truck operations and evaluate predefined charging and EU HoS-complient break strategies, including alternatives enabled by high-power charging technologies such as the Megawatt Charging System. While their study provides valuable insights into the interaction between charging technology and driver regulations, it only evaluates a limited number of predefined operating strategies and does not address the underlying scheduling problem.

1.2. Research gap and contributions

It remains unclear to what extent charging can be absorbed within mandatory driver breaks and daily rests in multi-day long-haul operations, and how incomplete synchronization affects route duration and schedule feasibility. Existing studies focus primarily on distribution networks and single-day operations, whereas

multi-day operations involve repeated charging, mandatory breaks, and recurring daily rests, creating greater potential for both synergies and conflicts between energy and regulatory requirements. Furthermore, existing synchronization studies rely primarily on mixed-integer formulations and heuristic solution approaches and do not provide an exact method for jointly optimizing charging, breaks, and daily rests over multiple days.

To address this gap, we study a multi-day variant of the Electric Vehicle Truck Driver Scheduling Problem (EVTDSP), which determines how charging activities, mandatory breaks, and daily rest periods should be scheduled and synchronized in long-haul battery-electric freight operations. Given a prescribed route and a set of charging opportunities, the objective is to minimize total route duration while satisfying both HoS and battery constraints.

The contributions of this paper are threefold. First, we extend the truck driver scheduling problem to the multi-day battery-electric setting by explicitly integrating charging decisions with both mandatory breaks and recurring daily rest periods. Second, we develop an exact resource-constrained shortest path formulation and labeling algorithm by extending the framework of Goel & Irnich (2017) with battery dynamics, tailored resource extension functions, and dominance rules. Third, we introduce an open-source benchmark set for the EVTDSP informed by interviews and operational data collected from long-distance trucking companies in Norway and Northern Europe, and use it to systematically analyze how route structure, charging conditions, and rest requirements affect schedule feasibility, charging-rest synchronization, and the value of partial recharging.

2. Problem definition

updated based on Magnus his new version. OK Now?

This paper studies the Electric Vehicle Truck Driver Scheduling Problem (EVTDSP), which concerns the construction of HoS-compliant and energy-feasible schedules for a battery-electric truck operating on a long-haul route. Given a predefined sequence of customer visits, the objective is to minimize the total route duration by determining the timing of mandatory breaks and daily rests, as well as when, where and how much to charge the battery. We consider a one-week planning horizon, $T^{\text{horizon}} = 168$ hours, reflecting the regulatory limit on the duration of a weekly work cycle.

The problem is defined on a directed graph $\mathcal{G} = (\mathcal{N}, \mathcal{A})$, where $\mathcal{N}$ and $\mathcal{A}$ denote the sets of nodes and arcs, respectively. The node set comprises the customer nodes $\mathcal{N}^C$, the charging-station nodes $\mathcal{N}^S$, the origin o, and the destination d, i.e., $\mathcal{N} = \mathcal{N}^C \cup \mathcal{N}^S \cup \{o, d\}$. Because the sequence of customer visits is predetermined and only charging stations located along this route are considered, all nodes have a unique order. Let $(i_1, \ldots, i_{|\mathcal{N}|})$ denote this order, with $i_1 = o$ and $i_{|\mathcal{N}|} = d$. The arc set connects consecutive nodes: $\mathcal{A} = \{(i_k, i_{k+1}) \mid k = 1, \ldots, |\mathcal{N}| - 1\}$. Figure 1 illustrates the graph for a route with three customers and six charging stations.

Each customer node $i \in \mathcal{N}^C$ requires a fixed, non-interruptible service time s_i. For each arc $(i, j) \in \mathcal{A}$, the distance d_{ij} and driving time t_{ij} are known. We assume a constant driving speed v and a constant energy-consumption rate α per unit of distance. The vehicle has battery capacity B, departs from the origin

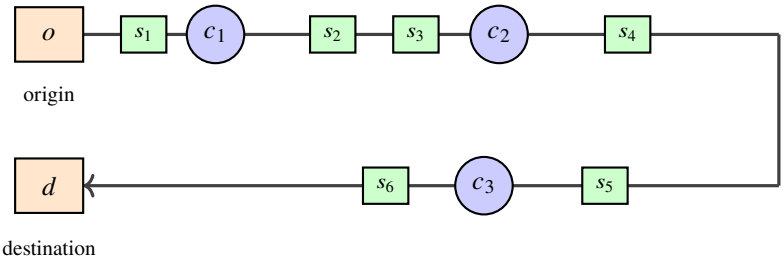


Figure 1: Illustrative route structure for the EVTDSP. The truck travels from origin o to destination d, visiting customers (c_1, c_2, c_3), with charging stations $(s_1, \ldots, s_6)$ available along the route.

fully charged, and must maintain a nonnegative state of charge throughout the route.

All charging stations are assumed to provide a constant charging power P^{ch}. Charging is therefore linear: adding q units of energy requires q/P^{ch} units of time. At a charging station, the vehicle may continue without charging or charge any amount up to the available battery capacity. Partial charging is thus part of the general problem definition, while full charging is obtained by restricting the vehicle to charge to capacity whenever a charging activity is selected. Multiple charging stops are allowed, but charging cannot be performed during customer service.

The driver is subject to the EU HoS rules represented by four limits: $t_{\texttt{drive}|B}$ and $t_{\texttt{elapsed}|B}$ bound the driving and elapsed working time, respectively, between consecutive breaks or rests, while $t_{\texttt{drive}|R}$ and $t_{\texttt{elapsed}|R}$ impose the corresponding limits between consecutive daily rests. A break lasts at least $t_{\texttt{break}}$, and a daily rest lasts at least $t_{\texttt{rest}}$. The driver is fully rested at the origin. Figure 2 illustrates a feasible schedule under these rules. We focus on standard 45-minute breaks and 11-hour daily rests; regulatory options such as split breaks and reduced daily rests are not considered.

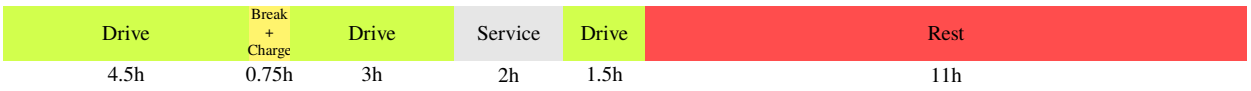


Figure 2: Example of an HoS-compliant schedule for a fully rested driver.

Breaks and daily rests may be taken at customer nodes, charging-station nodes, or intermediate points along an arc, but they cannot overlap with customer service. At charging stations, charging may be performed simultaneously with a break or daily rest. Only the part of a charging activity extending beyond the concurrent break or rest contributes additional time to the schedule.

The EVTDSP therefore determines a sequence of driving, service, charging, break, and rest activities that completes the prescribed route within T^{horizon}, satisfies all HoS and battery constraints, and minimizes the elapsed time from departure at o to arrival at d.

3. SPPRC Framework

- @mostafa+Magnus: resources elapsed since break or rest not interval valued -> Long discussion -> Not needed under our assumptins -> Explain
- @mostafa -> latest|R and latest|B -> only needed for time windows -> remove

The EVTDSP can be formulated as a shortest path problem with resource constraints (SPPRC) and solved using a labeling algorithm. We build on the labeling framework of Goel & Irnich (2017), extending it to jointly represent battery dynamics, charging decisions, and the synchronization of charging with mandatory breaks and daily rests. The prescribed route is transformed into a parameter-free activity network in which driving, breaks, daily rests, customer service, and charging are represented by activity arcs. Feasibility is tracked through temporal, Hours-of-Service, and battery resources propagated using resource extension functions (REFs).

The framework supports both full- and partial-recharge strategies. We focus on the more general partial-recharge formulation, in which the charging amount is a decision variable. The full-recharge formulation is obtained as a special case and is presented in Appendix A.

The main notation used throughout the remainder of the paper is summarized in Table 1. The remainder

Table 1: Main notation.

Symbol	Description
Networks and sets	
$\mathcal{G} = (\mathcal{N}, \mathcal{A})$	Prescribed route graph, with node set $\mathcal{N}$ and arc set $\mathcal{A}$.
$\widetilde{\mathcal{G}} = (\widetilde{\mathcal{N}}, \widetilde{\mathcal{A}})$	Parameter-free activity network used by the labeling algorithm.
$\mathcal{N}^C$, $\mathcal{N}^S \subseteq \mathcal{N}$	Sets of customer and charging-station nodes, respectively.
o, d	Origin and destination nodes.
Route and vehicle parameters	
d_{ij}, t_{ij}	Distance and driving time of route arc $(i, j) \in \mathcal{A}$.
s_i	Service time at customer node i.
v, α	Constant driving speed and energy-consumption rate per unit of distance.
B, P^{ch}	Battery capacity and charging power.
T^{horizon}	Planning horizon.
Hours-of-Service parameters	
t_{break}, t_{rest}	Minimum break and daily-rest durations.
$t_{\texttt{drive}\mid B}$, $t_{\texttt{elapsed}\mid B}$	Driving- and elapsed-time limits between consecutive breaks or rests.
$t_{\texttt{drive}\mid R}$, $t_{\texttt{elapsed}\mid R}$	Driving- and elapsed-time limits between consecutive daily rests.
Labeling notation	
$\mathcal{L} = (i, \pi, \mathbf{T})$	Label at node i, with predecessor information π and resource vector $\mathbf{T}$.
l_r, l'_r	Value of resource r before and after a resource extension.
Δ	Driving duration determined by a driving extension.
q	Energy added during a charging activity.
SOC	Battery state of charge.
f^{start}, f^{drive}, f^{break}, f^{rest}, f^{visit}, f^{charge}	Resource extension functions associated with the activity arcs.

of this section first describes the parameter-free network structure, followed by the label resources, REFs, and the feasibility and dominance rules used in the labeling algorithm.

3.1. Parameter-free activity network

We transform the route graph $\mathcal{G} = (\mathcal{N}, \mathcal{A})$ into a parameter-free network[1] $\mathcal{G} = (\widetilde{\mathcal{N}}, \widetilde{\mathcal{A}})$, on which the labeling algorithm is performed. The two graphs serve different purposes. The route graph $\mathcal{G}$ specifies the ordered sequence of route nodes and the associated travel parameters, whereas $\widetilde{\mathcal{G}}$ represents the feasible sequences of driver and vehicle activities along that route.

The parameter-free network is constructed by replacing each route arc $(i, j) \in \mathcal{A}$ with an activity subnetwork. Its node set $\widetilde{\mathcal{N}}$ consists of the original route nodes and two intermediate state nodes associated with each route arc: a fit state $\tilde{n}_{ij}^{\text{fit}}$ and a dull state $\tilde{n}_{ij}^{\text{dull}}$. Hence,

$$\widetilde{\mathcal{N}} := \mathcal{N} \cup \left\{ \tilde{n}_{ij}^{k} \mid (i, j) \in \mathcal{A},\ k \in \{\text{fit}, \text{dull}\} \right\}.$$

A fit state represents a situation in which driving may continue. A dull state represents a decision point reached after a driving extension. If the current route arc has not yet been completed, a break or daily rest must be taken before driving can resume; if the head node has been reached, the corresponding visit REF may instead be applied.

The arc set $\widetilde{\mathcal{A}}$ contains directed activity arcs connecting the route and intermediate nodes. Each activity arc represents a permissible driver or vehicle activity and is associated with an REF that updates the resources of the label being extended. Figure 3 illustrates the activity subnetworks generated by two consecutive route arcs, (n, s) and (s, m), where $s \in \mathcal{N}^S$ is a charging-station node.

The activity arcs in $\widetilde{\mathcal{A}}$ are: *start arcs*, which initialize the traversal of a route arc; *driving arcs*, which advance the vehicle along the current route arc; *break arcs* and *rest arcs*, which represent the corresponding HoS activities; *visit arcs*, which complete traversal of a route arc and process the route node reached, including customer service where applicable; and *charging arcs*, which determine the charging amount at a charging-station node and initialize traversal of the subsequent route arc. A charging amount of zero represents passing the station without charging.

3.2. Labeling algorithm

We solve the SPPRC using a labeling algorithm on the parameter-free activity network. The algorithm starts with an initial label at the origin o, representing a fully rested driver and a fully charged vehicle. At each iteration, an unprocessed label is extended along every feasible outgoing activity arc. The associated resource extension function (REF) updates the temporal, regulatory, and energy resources. Infeasible and dominated labels are discarded, and a complete schedule is obtained when a label reaches the destination d.

[1]The term *parameter-free* refers to the representation of activity durations. The network does not contain separate arcs for different possible durations of an activity. Instead, the duration is determined by the corresponding REF from the resources of the label being extended. In particular, the driving duration Δ is computed dynamically as the maximum feasible driving increment before the next route node is reached or an HoS limit becomes binding.

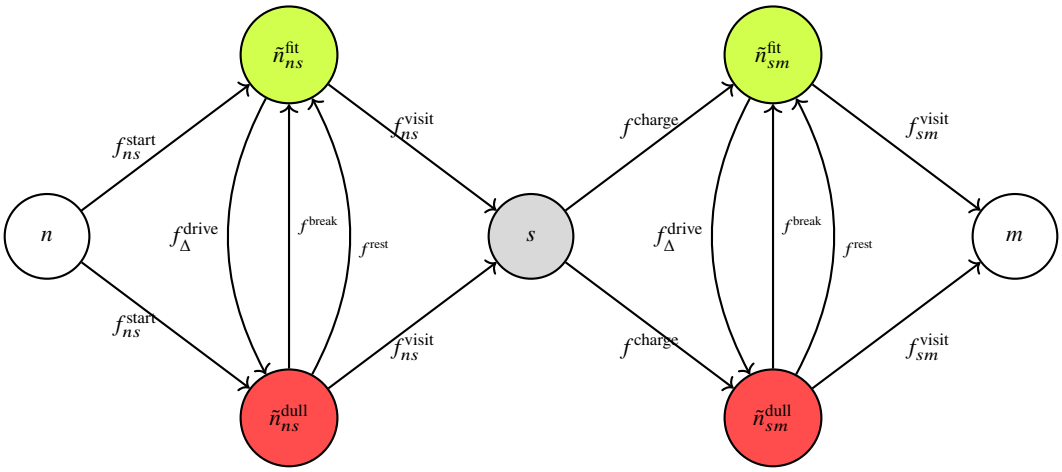


Figure 3: Parameter-free activity subnetworks for two consecutive customer nodes, n and m, with an intermediate charging-station node s. Each activity arc in $\widetilde{\mathcal{A}}$ is associated with the indicated resource extension function.

3.2.1. Label definition

A label represents a partial path in the parameter-free network $\widetilde{\mathcal{G}}$, together with the resource state needed to assess its feasibility and extend it into a complete schedule. Formally, $\mathcal{L} = (i, \pi, \mathbf{T})$, where $i \in \widetilde{\mathcal{N}}$ is the current node, π stores the predecessor information required to reconstruct the partial path, and $\mathbf{T}$ contains the associated resources.

The resource vector $\mathbf{T}$ tracks schedule time, progress along the current route arc, the HoS counters associated with breaks and daily rests, and the vehicle SOC. Table 2 summarizes the individual resources and their feasibility windows. Table 2 summarizes the resources and their feasibility windows.

Table 2: Label resources with corresponding feasibility windows.

Resource	**Type**	**Description**	**Feasibility Window**
`time`	Temporal	Total time elapsed since departure	$[0, T^{\text{horizon}}]$
`timeToNext`	Temporal	Remaining driving time on current route arc	$[0, t_{nm}]$
`drive|R`	Regulatory	Driving time since last daily rest	$[0, t_{\texttt{drive}\|R}]$
`drive|B`	Regulatory	Driving time since last break or rest	$[0, t_{\texttt{drive}\|B}]$
`elapsed|R`	Regulatory	Elapsed time since last daily rest	$[0, t_{\texttt{elapsed}\|R}]$
`elapsed|B`	Regulatory	Elapsed time since last break or rest	$[0, t_{\texttt{elapsed}\|B}]$
`SOC`	Energy	Battery state of charge	$[0,\ B]$

Under partial recharge, the continuous charging decision is represented through interval-valued time and SOC resources, while the HoS resources remain scalar-valued. The interpretation and propagation of these interval endpoints are detailed in Section 3.2.3. The resulting resource vector is

$$\mathbf{T} = \left(l^{\min}_{\texttt{time}}, l^{\max}_{\texttt{time}}, l_{\texttt{timeToNext}}, l_{\texttt{drive}|\texttt{R}}, l_{\texttt{drive}|\texttt{B}}, l_{\texttt{elapsed}|\texttt{R}}, l_{\texttt{elapsed}|\texttt{B}}, l^{\min}_{\texttt{SOC}}, l^{\max}_{\texttt{SOC}}\right).$$

3.2.2. Temporal and HoS resource extensions

Each activity arc $a =\in \widetilde{\mathcal{A}}$ is associated with a resource extension function f_a. Given a resource vector $\mathbf{T}$ at node $i \in \widetilde{\mathcal{N}}$, the REF returns an updated resource vector $\mathbf{T}' = f_a(\mathbf{T})$, computes valid lower bounds on its components, or declares the extension infeasible. Extending a label $\mathcal{L} = (i, \pi, \mathbf{T})$ therefore produces a label $\mathcal{L}' = (j, \pi', \mathbf{T}')$, where π' contains the updated predecessor information. We use l_r and l'_r to denote the value of resource r before and after an extension, respectively.

The activity arcs in $\widetilde{\mathcal{A}}$ use six types of REFs: f_{ij}^{start}, $f_{\Delta}^{\text{drive}}$, f^{break}, f^{rest}, f_{ij}^{visit}, and f_{ij}^{charge}. The start, driving, break, rest, and visit REFs propagate the temporal and HoS resources, while the driving REF additionally accounts for battery-energy consumption. Table 3 summarizes these extensions. For the interval-valued time and SOC resources, the listed updates apply to both endpoints.

Table 3: Resource updates for the start, driving, break, rest, and visit REFs. A dash indicates that the resource remains unchanged.

Resource	f_{ij}^{start}	$f_{\Delta}^{\text{drive}}$	f^{break}	f^{rest}	f_{ij}^{visit}
l'_{time}	—	$l_{\text{time}} + \Delta$	$l_{\text{time}} + t_{\text{break}}$	$l_{\text{time}} + t_{\text{rest}}$	$l_{\text{time}} + s_j$
$l'_{\text{timeToNext}}$	t_{ij}	$l_{\text{timeToNext}} - \Delta$	—	—	—
$l'_{\text{drive\|R}}$	—	$l_{\text{drive\|R}} + \Delta$	—	0	—
$l'_{\text{drive\|B}}$	—	$l_{\text{drive\|B}} + \Delta$	0	0	—
$l'_{\text{elapsed\|R}}$	—	$l_{\text{elapsed\|R}} + \Delta$	$l_{\text{elapsed\|R}} + t_{\text{break}}$	0	$l_{\text{elapsed\|R}} + s_j$
$l'_{\text{elapsed\|B}}$	—	$l_{\text{elapsed\|B}} + \Delta$	0	0	$l_{\text{elapsed\|B}} + s_j$
l'_{SOC}	—	$l_{\text{SOC}} - \alpha v \Delta$	—	—	—

For a route arc $(i, j) \in \mathcal{A}$, t_{ij} denotes the required driving time from route node i to route node j, and s_j denotes the service time at node j (but also index of a charging station), with $s_j = 0$ when $j \notin \mathcal{N}^C$.

The start REF initializes traversal of an arc (i, j) by setting $l'_{\text{timeToNext}} = t_{ij}$. A driving extension advances the schedule by a duration Δ, increases the applicable HoS counters, reduces the remaining driving time, and consumes $\alpha v \Delta$ units of battery energy. The driving duration is determined endogenously as

$$\Delta = \min\Big\{ l_{\text{timeToNext}}, t_{\texttt{drive|}B} - l_{\texttt{drive|B}}, t_{\texttt{drive|}R} - l_{\texttt{drive|R}}, \\ t_{\texttt{elapsed|}B} - l_{\texttt{elapsed|B}}, t_{\texttt{elapsed|}R} - l_{\texttt{elapsed|R}} \Big\}.$$

A driving extension therefore terminates when the head node of the current route arc is reached or when an HoS limit becomes binding. The driving REF additionally updates SOC. The extension is feasible if and only if $l_{\text{SOC}} - \alpha v \Delta \geq 0$.

The break and rest REFs add the corresponding interruption duration and reset the applicable HoS counters. A break resets $l_{\text{drive|B}}$ and $l_{\text{elapsed|B}}$, whereas a daily rest resets both the break- and rest-related resources. Finally, the visit REF completes traversal of route arc (i, j), records arrival at route node j, and adds s_j to the temporal and elapsed-time resources.

3.2.3. Charging resource extension

The charging REF f_{ij}^{charge} introduces the coupling between time and battery energy. We first describe this relationship for a scalar charging decision and then explain how the continuum of charging decisions is represented compactly within a label.

Consider a partial label arriving at charging station i at time t with state of charge l_{SOC}. We assume a linear charging process, such that the energy added is proportional to the charging duration. The vehicle may charge any amount $0 \le q \le B - l_{\text{SOC}}$, requiring a charging duration $\tau(q) = \frac{q}{P^{\text{ch}}}$.

Charging may take place simultaneously with a break or daily rest. In the activity network, this synchronization is represented by applying the corresponding break or rest REF immediately before the charging REF at the same charging-station location. The break or rest REF first adds the interruption duration and resets the relevant HoS resources. The charging REF then recognizes this reset state and treats the already-recorded interruption as a charging window. Thus, the break or rest duration is not added to the schedule a second time.

Let $w(\mathcal{L})$ denote the synchronization window identified from the current HoS state:

$$w(\mathcal{L}) = \begin{cases} t_{\text{rest}}, & \text{if the preceding activity is a daily rest,} \\ t_{\text{break}}, & \text{if the preceding activity is a break,} \\ 0, & \text{otherwise.} \end{cases}$$

Only the portion of the charging duration exceeding this window increases the route duration. The additional, unsynchronized charging time is therefore

$$\delta(q, \mathbf{T}) = \left[\frac{q}{P^{\text{ch}}} - w(\mathbf{T}) \right]^{+}.$$

The charging REF, f_{ij}^{charge}, then updates the time and SOC resources as:

$$l'_{\text{time}} = l_{\text{time}} + \delta(q, \mathbf{T}), \qquad l'_{\text{SOC}} = l_{\text{SOC}} + q, \qquad 0 \le q \le B - l_{\text{SOC}}.$$

The same unsynchronized duration is added to the applicable elapsed-time HoS resources, whereas the accumulated driving-time resources remain unchanged. After charging, `timeToNext` is initialized for the subsequent route arc.

Each feasible charging amount produces a paired departure-time and SOC value. Because charging is linear, the continuum of attainable combinations can be represented by the two endpoints $\left(l_{\text{time}}^{\min}, l_{\text{SOC}}^{\min}\right)$ and $\left(l_{\text{time}}^{\max}, l_{\text{SOC}}^{\max}\right)$. These endpoints define an attainable time–SOC line segment, rather than two independent intervals. The segment represents every feasible partial-charging decision between an earlier departure with less energy and a later departure with more energy.

Charging that fits within an available break or rest window increases SOC without delaying departure.

Consequently, charging less than the amount available at no additional time is not useful. If

$$\frac{B - l_{\mathrm{SOC}}}{P^{\mathrm{ch}}} \leq w(\mathbf{T}),$$

a full recharge can be completed entirely within the synchronized interruption. Otherwise, the label retains the trade-off between departing earlier with less energy and remaining longer to obtain a higher SOC.

The algorithm propagates this trade-off through subsequent driving, HoS, and visit extensions rather than fixing a charging amount immediately. Downstream energy requirements and HoS restrictions determine which portion of the represented interval can lead to a feasible and time-efficient completion. Dominance likewise compares the energy available at relevant times along the represented time–SOC segment. At the destination, the algorithm selects the feasible label with the shortest completion time. By tracing this label backward through the sequence of preceding labels, the algorithm determines the exact charging duration and energy added at each visited charging station.

3.2.4. Feasibility and dominance

clean up! After each resource extension, the resulting label must remain feasible. Feasibility is enforced through the resource windows in Table 2; labels violating any bound are discarded immediately. Under partial recharge, the bounds on `time` and `SOC` apply to both endpoints of the attainable time–SOC segment. The structure of $\widetilde{\mathcal{G}}$ additionally restricts the permissible extensions at each node. Driving arcs may be used only from a fit state. From a dull state, a break or daily rest must be completed before driving resumes. Once $l_{\texttt{timeToNext}} = 0$, a visit arc from either intermediate state completes traversal of the current route arc. At a charging-station route node, charging arcs initialize traversal of the subsequent route arc.

To limit the number of labels stored at each node in $\widetilde{\mathcal{N}}$, dominance is applied among labels ending at the same node. Let $\mathcal{L}_1 = (i, \pi_1, \mathbf{T}_1)$ and $\mathcal{L}_2 = (i, \pi_2, \mathbf{T}_2)$ be two feasible labels at $i \in \widetilde{\mathcal{N}}$. Label $\mathcal{L}_1$ dominates $\mathcal{L}_2$ if and only if:

$$\begin{aligned}
l^{(1)}_{\mathrm{timeToNext}} &\leq l^{(2)}_{\mathrm{timeToNext}}, \\
l^{(1)}_{\mathrm{drive|R}} &\leq l^{(2)}_{\mathrm{drive|R}}, & l^{(1)}_{\mathrm{drive|B}} &\leq l^{(2)}_{\mathrm{drive|B}}, \\
l^{(1)}_{\mathrm{elapsed|R}} &\leq l^{(2)}_{\mathrm{elapsed|R}}, & l^{(1)}_{\mathrm{elapsed|B}} &\leq l^{(2)}_{\mathrm{elapsed|B}}, \\
l^{(1)}_{\mathrm{latest|R}} &\geq l^{(2)}_{\mathrm{latest|R}}, & l^{(1)}_{\mathrm{latest|B}} &\geq l^{(2)}_{\mathrm{latest|B}}, \\
l^{\mathrm{min},(1)}_{\mathrm{time}} &\leq l^{\mathrm{min},(2)}_{\mathrm{time}}, \\
l^{\mathrm{min},(1)}_{\mathrm{SOC}} &\geq l^{\mathrm{min},(2)}_{\mathrm{SOC}}, & l^{\mathrm{max},(1)}_{\mathrm{SOC}} &\geq l^{\mathrm{max},(2)}_{\mathrm{SOC}}.
\end{aligned}$$

and, at the earliest departure time of label 2, label 1 must offer at least the same battery energy, accounting for optional charging between $l^{\mathrm{min},(1)}_{\mathrm{time}}$ and $l^{\mathrm{max},(1)}_{\mathrm{time}}$. Since the objective is to minimize total route duration, a label is preferable if its resource values are at least as good for continuing the route. For the scalar HoS resources (`timeToNext`, `drive|R`, `drive|B`, `elapsed|R`, and `elapsed|B`), which are monotone

non-decreasing along every REF except break, rest, and synchronized charging, label 1 dominates label 2 at the same node if each of these resources is no larger in label 1 than in label 2. This pairwise comparison is sufficient for SPPRC with monotone resources and preserves optimality (Irnich & Desaulniers, 2005; Goel & Irnich, 2017). Resources `latest|R` and `latest|B` are compared in the reverse direction because larger values correspond to greater remaining scheduling flexibility. The partial-recharge extension introduces interval-valued time and SOC resources. The additional dominance conditions on $l^{\min}_{\text{time}}$, $l^{\min}_{\text{SOC}}$, $l^{\max}_{\text{SOC}}$, and the battery level at $t = l^{\min,(2)}_{\text{time}}$ follow the interval dominance rule for delivery paths with optional recharging (Desaulniers et al., 2016).

4. Experimental design

This section outlines the experimental design used in the computational study. We first describe the benchmark instances used to evaluate the proposed framework. We then introduce a baseline scheduling heuristic, which serves as a reference for assessing the benefits of jointly optimizing charging, breaks, and daily rests.

4.1. Benchmark instances

- SB: create new figure with all customers.

Long-haul freight operations vary substantially in route length, customer distribution, and operating conditions. These characteristics influence the potential gains from jointly scheduling charging, mandatory breaks, and daily rests. We therefore develop a diverse set of structural route instances and evaluate them under different charging-station coverage levels, battery capacities, charging powers, and energy-consumption rates. Rather than reproducing individual observed routes, the instances represent a broad range of realistic long-haul freight operations and constitute, to the best of our knowledge, the first benchmark set for the long-haul EVTDSP.

The benchmark design was informed by interviews with freight operators in Scandinavia and Northern Europe. With assistance from the Norwegian Trucking Association (NLF), 28 companies from different regions and transport segments were contacted, of which eight participated. They represented general freight, container, seafood, temperature-controlled, waste, and bulk transport. The interviews highlighted considerable variation in route length, route duration, number of stops, and customer distribution. Customers were often geographically clustered, resulting in routes that combine long intercity movements with shorter local collection and distribution activities. The interviews also emphasized the importance of charging-station availability and the potential gains from coordinating charging with mandatory driver breaks and daily rests.

Based on these industry inputs, we develop an instance generator for heterogeneous long-haul routes. The generator first defines route instances along three main dimensions: route length, customer distribution, and the number of customer stops. Scenario parameters are added subsequently to vary charging-station

spacing, battery capacity, charging power, and energy consumption without changing the underlying route and customer structure.

Routes are classified as short, medium, or long, corresponding to increasing route distances. Specifically, short routes span 800–1200 km and typically require 1–2 overnight rests, while medium routes span 1500–2500 km and require 3–4 overnight rests. Long routes span 3000–4000 km and require 5–6 overnight rests, reaching the maximum multi-day operating horizon of six days before a longer weekly rest becomes necessary under the HoS regulations. For each instance, the total route length D_{total} is sampled uniformly from the interval associated with its route class.

Each route contains between one and three industrial clusters, interpreted as groups of geographically proximate customers. Cluster centers are positioned along intermediate sections of the route, so that customer visits are concentrated around these clusters and the first and last long-haul legs are typically the longest.

Customer stops are generated in two steps. First, one of three customer-count categories is selected: *few* (1–3 customers), *medium* (4–5 customers), or *many* (6–15 customers), and the total number of customers N is sampled from the corresponding range. Second, customers are allocated among the route clusters, with at least one customer assigned to each cluster. In the base configurations, customers are distributed approximately evenly among clusters, refered to as balanced instances. For selected multi-cluster configurations, an unbalanced variant is also generated by assigning most customers to one randomly selected dominant cluster and only a few to the remaining clusters. For example, 12 customers across three clusters may be allocated as $[4, 4, 4]$ in a balanced instance and $[9, 2, 1]$ in an unbalanced instance.

Customer locations are then placed within 75 km of their cluster center in either direction along the route, while maintaining local customer-to-customer distances of 30–150 km. This produces long intercity movements between clusters and shorter local movements within them. Service times are linked to the customer-count category: 20–30 minutes per stop for few-customer instances, 10–20 minutes for medium instances, and 5–10 minutes for many-customer instances. Instance names indicate the route class, number of clusters, number of customers, and, where applicable, the allocation variant; for example, `long_cl2_cu5_Balanced`.

A total of 42 unique structural benchmark instances are generated by combining the route, cluster, and customer classes described above. Table 4 summarizes the structural design, while the full set of instances and their characteristics is reported in Table C.11.

Figure 4 illustrates three representative instances. Charging stations are shown using the base-case maximum spacing of 50 km; the spacing parameter is defined below.

The computational experiments use different technological and infrastructure parameters, as summarized in Table 5.

The selected battery capacities of 350, 500, and 750 kWh span current and announced heavy-duty electric-truck configurations, which range from approximately 360 to 780 kWh (Winkler et al., 2026; Volvo Trucks, 2025b; Mercedes-Benz Trucks, 2025; Volvo Trucks, 2025a). Charging powers of 350 and 1000 kW represent high-power CCS charging and prospective Megawatt Charging System capabilities, respectively. For example, the Mercedes-Benz eActros 600 supports up to 400 kW with CCS and 1000 kW with MCS

Table 4: Structural design of the benchmark instances.

Route characteristics	**Short**	**Medium**	**Long**
Route length	800–1200 km	1500–2500 km	3000–4000 km
Overnight rests	1–2	3–4	5–6
Sampled customers	2–12	2–13	2–14
Customer characteristic	**Few**	**Medium**	**Many**
Number of customers	1–3	4–5	6–15
Service time per stop	20–30 min	10–20 min	5–10 min
Cluster characteristics			
Clusters per route	1–3 clusters		
Customer distribution across clusters	Balanced; unbalanced		

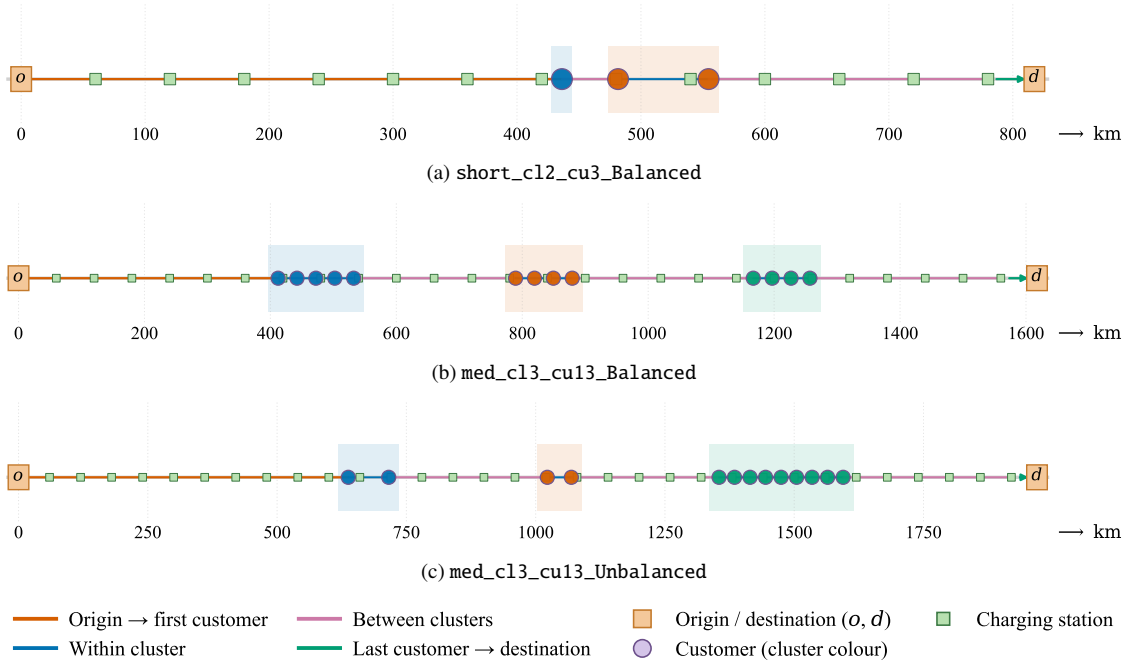


Figure 4: Three representative benchmark instances illustrating variation in route length and customer distribution. Charging stations are shown using a maximum spacing of 50 km. colors of some arcs are a bit weird -> check steffen

Table 5: Scenario parameters and fixed HoS parameters.

Scenario parameters			
Parameter	**Symbol**	**Base value**	**Sensitivity values**
Battery capacity	B	500 kWh	350, 750 kWh
Charging power	P_{ch}	350 kW	1000 kW
Maximum charging-station spacing	d_{CS}	50 km	40, 60, . . . , 100 km
Energy consumption	α	1.49 kWh/km	1.12, 2.08 kWh/km
Fixed HoS parameters			
Parameter	**Symbol**	**Value**	
Maximum continuous driving time	$t_{\text{drive}\|B}$	4.5 h	
Daily driving limit	$t_{\text{drive}\|R}$	9 h	
Maximum elapsed time before break	$t_{\text{elapsed}\|B}$	6 h	
Maximum elapsed time before daily rest	$t_{\text{elapsed}\|R}$	13 h	
Minimum break duration	t_{break}	0.75 h	
Minimum daily rest duration	t_{rest}	11 h	

(Mercedes-Benz Trucks, 2025). In our model, these values are treated as constant charging rates; the 1000 kW case should therefore be interpreted as an idealized near-future scenario. Energy-consumption rates of 1.12, 1.49, and 2.08 kWh/km represent operating conditions ranging from favorable to demanding, around values reported in empirical studies and real-world trials (Volvo Trucks, 2025b; Le Corguillé et al., 2025). Travel times are calculated using a fixed average speed of 60 km/h, based on historical data provided by an industry partner.

The maximum charging-station spacing ranges from 40 to 100 km. This range reflects the 2030 targets in EU Regulation 2023/1804 (AFIR): publicly accessible charging sites for heavy-duty vehicles must be available in each direction of travel at intervals of at most 60 km along the TEN-T core network and 100 km along the comprehensive network; including at least two 350 kW charging points on the core network and one on the comprehensive network (European Parliament and Council of the European Union, 2023).

Charging stations are inserted separately along each route leg. If a leg is shorter than the selected maximum spacing, one station is placed at its midpoint. Otherwise, stations are distributed such that the distance between consecutive charging opportunities does not exceed the selected spacing. The spacing parameter thus represents an upper bound rather than the exact distance between neighboring stations.

The HoS parameters are fixed across all experiments. Following Goel & Irnich (2017), we use maximum continuous and daily driving times of 4.5 and 9 h, maximum elapsed times of 6 and 13 h before a break and daily rest, and minimum break and daily-rest durations of 0.75 and 11 h, respectively.

4.2. Heuristic benchmark(s)

Suggestions from Steffen below, please review. Let's test both versions, and then decide whether to include 0, 1 or 2 heuristic comparisons in the final paper.

To evaluate how much of the benefit of joint optimization can be obtained using simple operational rules, we consider two heuristics. The first mimics the sequential decisions that could be made by a driver along the route, whereas the second retains the labeling framework but restricts the set of decisions considered.

Both heuristics use a synchronization parameter $t^{\text{sync}} \geq 0$, which represents how early a driver is willing to take a break or daily rest in order to synchronize it with charging. Specifically, when a charging station is reached, an upcoming break or daily rest may be taken at that station if it would otherwise become mandatory within the next t^{sync} time units. Thus, $t^{\text{sync}} = 0$ corresponds to a reactive policy, whereas larger values allow synchronization. Let's perform a sensitivity analysis on this for $t_{\text{synch}} \in \{0, 0.5, 1.0, 1.5, 2.0\}$ (hours).

4.2.1. Driver-inspired sequential heuristic

The driver-inspired heuristic constructs a feasible schedule sequentially along the prescribed route using only the current vehicle and HoS state and information about the next charging opportunities. The heuristic is based on the principle of driving as far as reasonably possible and combining charging with mandatory driver interruptions whenever possible.

The following rules are applied:

1. *Drive as far as possible.* When a charging station is reached, the vehicle continues without charging if the available SOC is sufficient to reach the next charging opportunity or the destination, unless the station provides a synchronization opportunity as defined below.
2. *Synchronize upcoming HoS interruptions.* Upon arriving at a charging station, if a break or daily rest will become mandatory within t^{sync}, the interruption is taken early at that station and synchronized with charging. If both are eligible, the daily rest is preferred.
3. *Combine necessary charging with an HoS interruption.* If the available SOC is insufficient to reach the next charging opportunity, the vehicle charges at the current station. Every such charging stop is combined with a break or daily rest. A daily rest is taken if it becomes due within t^{sync}; otherwise, a break is taken.
4. *Take unavoidable HoS interruptions when required.* If an HoS limit becomes binding and no charging station has been reached within the preceding t^{sync} window, the required break or daily rest is taken at that point. The vehicle then continues along the route.

We consider two simple charging policies within this heuristic. Under the *full-charge policy*, the vehicle charges to full capacity whenever a charging stop is made. Under the *overlap-only policy*, charging is performed only during the concurrent break or daily rest, such that the charging activity itself does not extend the stop. If this charged amount is insufficient to reach the next charging opportunity, charging is extended only as much as required to restore reachability.

4.3. Baseline heuristic: driver-inspired sequential scheduling

To provide an interpretable benchmark, we develop a driver-inspired sequential heuristic. The heuristic constructs a feasible schedule along the prescribed route without enumerating alternative partial paths. It

processes route nodes and charging opportunities in their travel order while maintaining the current time, state of charge (SOC), and HoS resources. Its decisions follow four rules.

First, the vehicle drives as far as possible. At a charging station, it passes without stopping when its SOC is sufficient to reach the next charging opportunity or the destination, unless an HoS interruption should be synchronized there. Second, if a break or daily rest will become mandatory within a synchronization horizon t_{sync}, the interruption is taken early at the current charging station and combined with charging. A daily rest takes priority when both interruptions are relevant. Third, if the available SOC is insufficient to reach the next charging opportunity, charging is performed at the current station and synchronized with a daily rest when one is due within t_{sync}, or otherwise with a break. Finally, if an HoS limit becomes binding while traveling between consecutive stops, the required interruption is taken on the road without charging.

The synchronization horizon controls the foresight of the heuristic. Its default value is

$$t_{\text{sync}} = \frac{d_{\text{CS}}}{v},$$

where d_{CS} is the charging-station spacing and v is the driving speed. Thus, at the base setting $d_{\text{CS}} = 60$ km and $v = 60$ km/h, $t_{\text{sync}} = 1$ h. An interruption is moved to the current charging station when the next station cannot be reached before the corresponding HoS limit becomes binding. Smaller values make the heuristic more reactive, whereas larger values permit earlier synchronization.

We evaluate two charging policies. Under the *full-charge* policy, each charging stop restores the battery to capacity. Under the *overlap-only* policy, charging is initially limited to the duration of the concurrent break or daily rest. Charging is extended beyond that interruption only when the resulting SOC would otherwise be insufficient to reach the next charging opportunity. Consequently, the overlap-only policy avoids unnecessary unsynchronized charging while preserving route feasibility. Two feasibility safeguards are included. A break is replaced by a daily rest when taking the break would leave insufficient daily elapsed time for a feasible continuation. Similarly, a break-synchronized charge is upgraded to a daily-rest-synchronized charge when its unsynchronized excess would violate the daily elapsed-time limit. After construction, an independent validator replays the complete activity timeline and verifies all battery and HoS constraints.

The synchronization-horizon sensitivity analysis in Figure 5 exhibits a U-shaped pattern for both driver-heuristic configurations. Mean route duration is minimized near $t_{\text{sync}} = 0.75$ h and remains comparatively low at 1 h. Shorter horizons miss useful charging–interruption synchronization opportunities, whereas longer horizons trigger interruptions prematurely. We therefore use $t_{\text{sync}} = 1$ h as an operationally interpretable default, corresponding to one charging-station spacing under the base configuration.

5. Computational results

Potential structure for new results

1. **Base case results (optional).**
    - one table summarizing the different approaches in the base case.

Algorithm 1 Driver-inspired sequential heuristic

Require: Prescribed route, battery capacity B, charging power P_{ch}, consumption rate α, CS spacing d_{CS}, synchronization horizon t_{sync}, policy $p \in \{\textsc{Full}, \textsc{Overlap}\}$

Ensure: Feasible activity schedule or an infeasibility declaration

Insert the charging opportunities along the prescribed route
Initialize time and HoS counters to zero and SOC $\leftarrow B$
for each next route node or charging opportunity j **do**
    while j has not been reached **do**
        Set Δ to the minimum of the remaining travel time, time to the next mandatory break, and time to the next daily rest
        Drive for Δ; update time, SOC, and HoS counters
        if an HoS limit binds before reaching j **then**
            Take a daily rest if required; otherwise take a break ▷ Rule 4
        end if
    end while
    if j is a charging station **then**
        Compute energy E_{next} needed to reach the next charging opportunity or destination
        Compute times t_B and t_R until a break and daily rest become mandatory
        if SOC $< E_{\text{next}}$ **then**
            Select a daily rest if $t_R < t_{\text{sync}}$; otherwise select a break
            Mark the stop as necessary charging ▷ Rule 3
        else if $t_R < t_{\text{sync}}$ or $t_B < t_{\text{sync}}$ **then**
            Select a daily rest if $t_R < t_{\text{sync}}$; otherwise select a break
            Mark the stop as early synchronization ▷ Rule 2
        else
            Pass the charging station without stopping ▷ Rule 1
            continue
        end if
        if $p = \textsc{Full}$ **then**
            Set target SOC to B
        else
            Set target SOC to the energy obtainable during the selected interruption
            Increase target SOC to E_{next} if required
        end if
        if a break and its excess charging time violate the daily elapsed limit **then**
            Replace the break by a daily rest
        end if
        Perform charging and the selected interruption concurrently
    else
        if service at j would violate an HoS elapsed-time limit **then**
            Take the required break or daily rest
        end if
        Perform service at j
    end if
end for
Validate the complete activity timeline independently
return schedule if valid; otherwise declare infeasibility

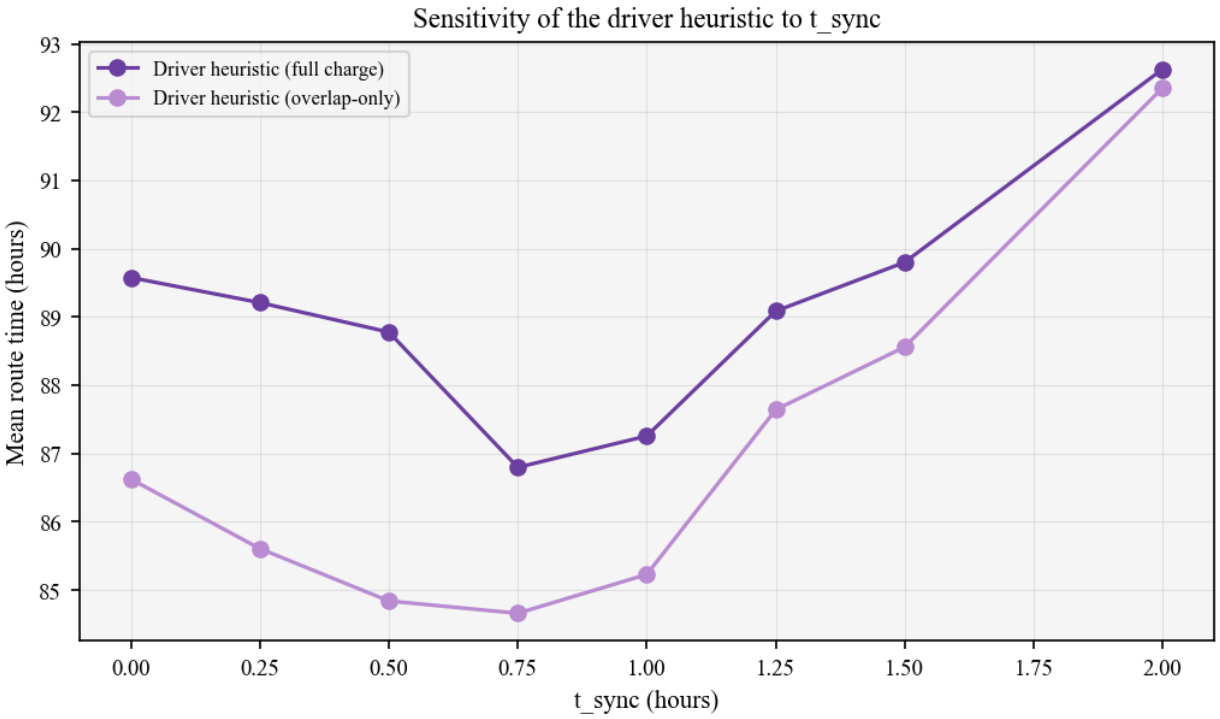


Figure 5: Sensitivity of mean route duration to the synchronization horizon t_{sync} for the full-charge and overlap-only driver heuristics. Horizontal lines indicate the mean route durations of the benchmark policies, while the vertical dotted line marks the default setting $t_{\text{sync}} = 1$ h.

- keep extremely short, or not include at all.

2. **Competitiveness of (optimized) electric operation**. Can smart electric charging compete with diesel, and under what conditions?
   - Smart (partial) charging vs diesel
   - Why/when EV can match diesel
   - Sensitivity: battery, charging power, consumption, station spacing
   - Feasibility
3. **The value of partial charging**. Does partial charging actually matter, and when?
   - Partial vs full in base case
   - How large is the gain?
   - Route length / difficult instances
   - WHY: synchronization, idle charging, number/location of stops
   - Sensitivity: under what technological conditions is partial charging valuable?
4. **Performance of heuristic scheduling strategies**. How much complex optimization is needed?

- Exact vs driver heuristic vs restricted labeling
- Sensitivity to t^{sync}
- Full-charge vs overlap-only heuristic

This section reports the computational results. We first compare partial recharge, full recharge, the greedy heuristic, and the diesel proxy in the base-case setting. We then examine how these differences vary across instance classes and relate them to charging synchronization. Finally, we analyze the sensitivity of the results to battery characteristics, charging infrastructure, and energy consumption.

5.1. Performance comparison of charging strategies

Table 6 summarizes the aggregate benchmark comparison for the base-case scenario. The results indicate that the dominant performance effect comes from optimization itself: schedules produced by the exact methods substantially outperform the greedy benchmark, highlighting the benefit of jointly optimizing charging decisions and driver scheduling under the Hours-of-Service regulation. Within this optimized setting, partial recharge provides a further systematic improvement over full recharge, indicating that additional value comes from adjusting charging amounts more precisely to downstream scheduling opportunities.

Table 6: Base-case comparison: partial recharge vs. alternative solution approaches.

Alternative	n	Mean route time (h)		Partial advantage (benchmark − partial)			
		Partial	Alternative	Mean (%)	Mean (min)	Min (min)	Max (min)
Full recharge	43	81.93	82.43	0.61	29.95	4.80	89.39
Greedy	42	83.14	119.15	28.83	2160.35	56.07	4225.01
Diesel proxy	43	81.93	81.30	−0.71	−37.55	−180.00	0.00

Each row compares partial recharge with one alternative on the same test instances (n). Partial advantage: positive values mean partial recharge is faster. Min/Max (min) are the smallest and largest instance-level time differences in minutes.

At the same time, the small average gap to the diesel reference case suggests that, when charging is coordinated effectively with mandatory breaks and daily rests, battery-electric operation can come close to diesel-like performance in the base-case setting. This average result is not uniform across instances, however. Figure 6 shows that the advantage of partial recharge over full recharge varies across the test set, and that the remaining EV–diesel gap is concentrated in a limited number of more difficult cases. The next subsection therefore examines where these gains are largest and how they relate to route structure.

The benefit of partial recharging increases in absolute terms with route length, although the percentage improvement over full recharging remains modest across all route classes. For short routes, partial recharging reduces the mean route time from 34.32 h to 34.12 h, corresponding to an average saving of 12 min. The mean saving increases to 27 min for medium routes and to around 50 min for long routes. Thus, the additional flexibility of partial recharging becomes more valuable as route duration and the number of charging decisions increase.

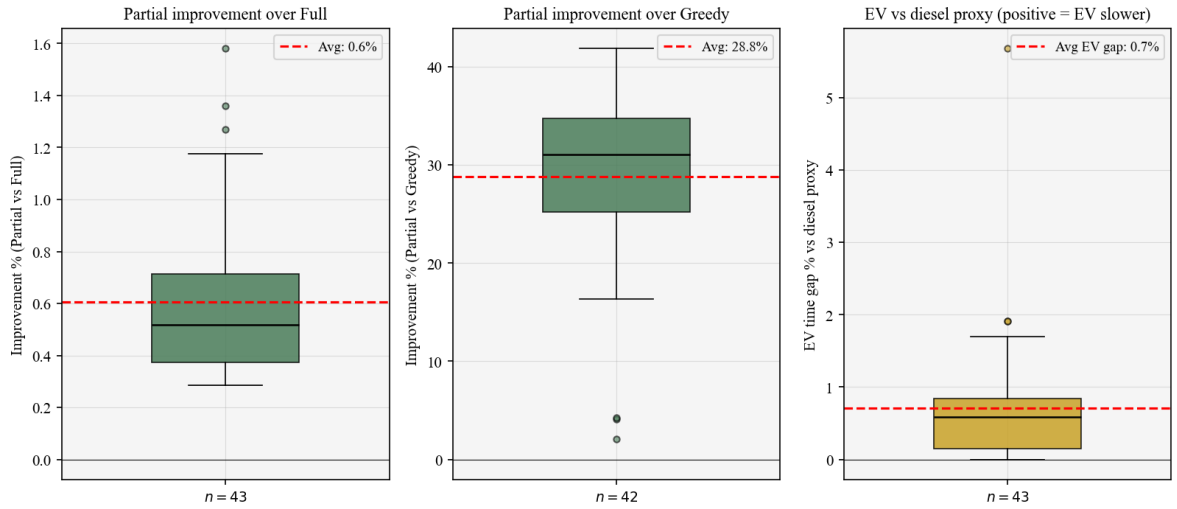


Figure 6: Per-instance comparison of partial recharge against full recharge, the greedy heuristic, and the diesel proxy in the base-case setting.

This pattern is also visible in the instances with the largest full-versus-partial gaps. The five strongest improvements all occur on long routes, namely L_cl3_cu4_U, L_cl3_cu3_B, L_cl2_cu2_B, L_cl2_cu2_U, and L_cl2_cu3_B, with savings ranging from 62.40 min to 89.39 min. These cases are precisely those in which longer routes create many charging decisions and therefore more scope to improve the schedule by reallocating charging and aligning it with mandatory interruptions. For example, in L_cl3_cu4_U, partial recharge reduces route time by 89.39 min, raises the synchronization share from 58.33% to 78.57%, reduces the number of unsynchronized charging events from 5 to 3, and cuts idle charging time from 122.76 min to 45.98 min, despite increasing the number of charging stops from 12 to 14. By contrast, the smallest gains occur on short routes S_cl3_cu4_U, S_cl2_cu4_U, S_cl2_cu3_B, S_cl3_cu5_B, and S_cl3_cu4_B where the savings are only 4.80 min to 9.60 min. In S_cl3_cu4_U, both strategies already achieve full synchronization, with zero unsynchronized charging and zero idle charging time, so the scope for further improvement is minimal. Detailed per-instance route times and charging-activity statistics are reported in Table D.13.

These results show that the operational value of partial recharge increases with route length and appears to be closely linked to charging synchronization. The next subsection examines this mechanism more directly by relating the observed time savings to the alignment of charging with breaks and daily rests.

5.2. The role of charging synchronization

The route-class results suggest that the value of partial recharge comes not from charging less, but from charging more effectively. Table 7 makes this mechanism explicit. On average, partial recharge uses more charging stops than full recharge (7.53 versus 5.30), yet still achieves a lower mean route time. The improvement arises because these additional charging stops are better aligned with mandatory driver interruptions. In particular, the synchronization share increases from 74.56% under full recharge to 81.89%

under partial recharge, while the number of unsynchronized charging events falls slightly from 1.58 to 1.42 despite the larger number of total charges. The main difference comes from break synchronization: partial recharge performs, on average, 2.58 charging stops during breaks, compared with only 0.19 under full recharge, whereas synchronization with daily rests remains essentially unchanged. The same table also clarifies why the greedy benchmark performs poorly. Although the greedy heuristic averages only 4.14 charging stops, none of them are synchronized with breaks or rests, so every charging event adds directly to route duration. The principal benefit of optimization is therefore not simply to reduce the amount of charging, but to embed charging more effectively within time that the driver is required to spend off the road anyway.

Table 7: Charging-synchronization statistics by solution approach in the base-case setting.

Approach	Route time (h)	Charges	Sync. share (%)	Sync. charges	Unsync. charges	Break-sync	Rest-sync
Full recharge	82.43	5.30	74.56	3.72	1.58	0.19	3.53
Partial recharge	81.93	7.53	81.89	6.12	1.42	2.58	3.53
Greedy heuristic	115.90	4.14	0.00	0.00	4.14	0.00	0.00

Entries report averages across the available base-case schedules for each approach. "Sync. charges" and "Unsync. charges" denote charging events that do and do not overlap with mandatory breaks or rests, respectively. "Break-sync" and "Rest-sync" report the average numbers of charging events synchronized with breaks and daily rests.

The same mechanism explains why the gains from partial recharge are largest on long routes. Table 8 compares the five instances with the largest and smallest reductions in route time relative to full recharge. In the largest-gain cases, all of which are long routes, full recharge is associated with comparatively low synchronization and substantial idle charging time. Partial recharge raises the synchronization share from 57.47% to 68.90%, reduces unsynchronized charging from 5.0 to 4.2 events on average, and cuts idle charging time from 94.41 min to 66.60 min, despite using slightly more charging stops. By contrast, the smallest-gain cases are all short routes. There, both strategies already achieve near-complete synchronization, virtually no unsynchronized charging, and negligible idle charging time, leaving little scope for further improvement. Thus, the advantage of partial recharge comes not from fewer charging stops, but from greater flexibility in allocating charging across stops so as to reduce idle and unsynchronized charging when routes become long enough for these effects to matter.

Table 8: Comparison of instances with the largest and smallest partial-recharge gains over full recharge.

Group	Class	Saving (min)	Sync. share (F/P)	Unsync. (F/P)	Idle (F/P)	CS stops (F/P)
Largest gains	Long	74.49	57.47/68.90	5.0/4.2	94.41/66.60	11.6/13.6
Smallest gains	Short	6.72	90.00/93.33	0.2/0.2	3.07/3.07	1.4/2.6

Entries report averages over the five instances with the largest and smallest values of *full recharge route time minus partial recharge route time*. Positive savings therefore indicate that partial recharge is faster. In columns marked "F/P", values are reported as *Full / Partial*.

The cases closest to the diesel benchmark reinforce this interpretation. In the six base-case instances

for which the optimized electric schedule matches the diesel benchmark exactly in total route time, every charging stop is synchronized with a mandatory break or rest, and both unsynchronized charging and idle charging are zero. Near-diesel performance is therefore achieved not by avoiding charging altogether, but by absorbing charging fully within the Hours-of-Service structure. Detailed per-instance evidence for these near-diesel cases is reported in Appendix D Table D.12.

5.3. Effects of battery and charging infrastructure characteristics

We next examine how technology and charging-infrastructure assumptions affect the ability of optimized electric-truck schedules to approach the diesel proxy. In this analysis, the benchmark is the full-recharge electric schedule, and the performance measure is the paired route-time gap relative to the diesel proxy:

$$g_{i,\theta} = 100 * \frac{T_{i,\theta}^{\text{full}} - T_i^{\text{diesel}}}{T_i^{\text{diesel}}},$$

where $T_{i,\theta}^{\text{full}}$ denotes the optimized full-recharge route time for instance i under parameter setting θ, and T_i^{diesel} denotes the corresponding diesel-proxy route time for the same route. Reported averages are computed over EV routes that remain feasible, while feasibility rates are reported separately. Figure 7 shows that EV-diesel gap is increasing when energy demand is high, battery capacity is low, or charging is slow. As the energy-consumption rate increases from 1.12 kWh/km to 2.08 kWh/km, the mean route-time penalty rises from 0.49 % to 3.05 %, reflecting the additional charging burden and reduced flexibility for absorbing recharging within mandatory breaks and daily rests. The opposite pattern is observed for battery capacity and charging power. Increasing battery capacity from 750 kWh to 1000 kWh reduces the mean penalty from 2.52 % to 1.04 %, while increasing charging power from 350 kW to 1000 kW reduces it from 2.73 % to 0.97 %. These reductions indicate that larger batteries and faster charging make it easier to complete required charging within off-duty periods.

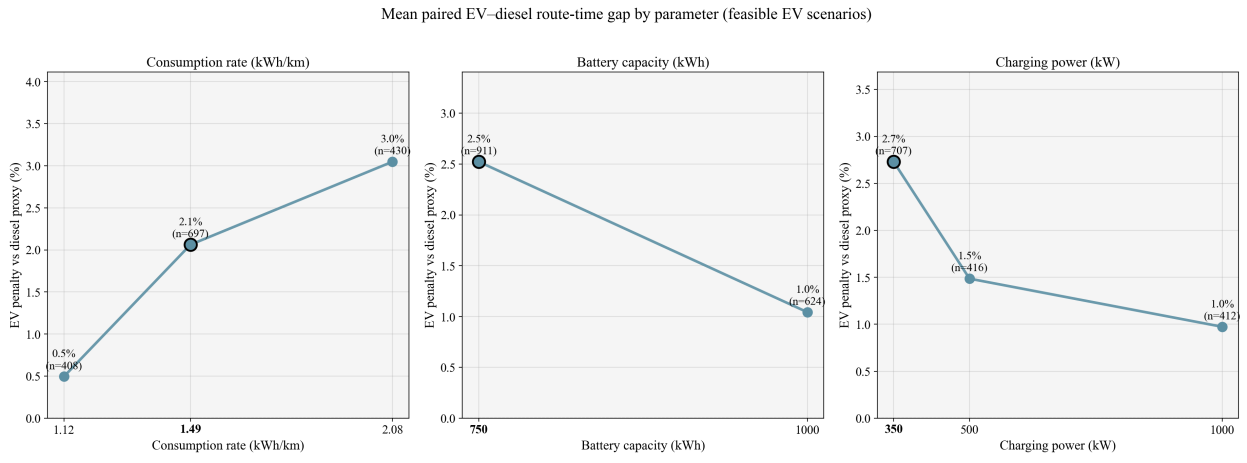


Figure 7: Mean paired route-time penalty of optimized full-recharge electric schedules relative to the diesel proxy, by technology parameter. The highlighted marker indicates the base-case parameter value used in the benchmark comparison.

The charging-pattern indicators in Table 9 support this interpretation. Higher consumption rates and smaller batteries increase the number of charging stops and the number of synchronized charging events. For example, the average number of charging stops rises from 1.44 at 1.12 kWh/km to 3.70 at 2.08 kWh/km. Similarly, the 750 kWh battery case requires 3.60 charging stops on average, compared with 1.94 for the 1000 kWh case. Charging power affects the mechanism differently: higher power reduces the time cost of each charging event, so even when charging remains necessary, the route-time penalty relative to diesel declines substantially.

Table 9: Sensitivity summary for optimized full-recharge schedules relative to the diesel proxy.

Parameter	Value	Gap (%)	Gap (min)	Feas. (%)	Charges	Sync.	Unsync.	Break	Rest
Consumption rate	1.12	0.49	13.17	100.0	1.44	1.32	0.13	0.24	1.08
(kWh/km)	1.49	2.06	83.85	99.2	3.31	2.95	0.36	1.2	1.75
	2.08	3.05	75.66	100.0	3.7	3.08	0.63	1.49	1.59
Station spacing	60	1.59	33.81	100.0	2.44	1.81	0.62	0.31	1.5
(km)	80	1.97	42.33	100	2.5	1.94	0.56	0.39	1.56
	100	2.31	50.07	100	2.61	2.06	0.56	0.56	1.5
	120	2.38	108.23	100	4.75	3.88	0.88	1.09	2.78
	140	3.46	171.73	100	5.28	4.78	0.5	2.59	2.19
	160	3.16	141.65	100	4.88	4.47	0.41	1.97	2.5
	180	3.46	166.02	100	5.39	5.24	0.15	2.58	2.67
	200	4.04	196.23	100	5.24	5.06	0.18	3.12	1.94
	220	3.9	186.43	100	5.33	5.06	0.27	2.97	2.09
	240	3.64	167.42	100	4.88	4.62	0.25	2.22	2.41
	260	3.72	183.44	97.0	5.55	5.32	0.23	3.03	2.29
	300	4.95	235.76	84.8	4.48	4.37	0.11	2.7	1.67
Battery capacity	750	2.52	88.59	99.4	3.6	3.21	0.39	1.55	1.66
(kWh)	1000	1.04	25.08	100	1.94	1.59	0.35	0.27	1.32
Charging power	350	2.73	100.86	99.2	3.54	3.23	0.3	1.43	1.8
(kW)	500	1.49	36.47	100	2.46	2.13	0.33	0.73	1.4
	1000	0.97	23.97	100	2.34	1.81	0.53	0.64	1.17

Gap (% and min) is the mean paired route-time penalty of full-recharge vs. diesel-proxy schedules. Feas. reports full-recharge feasibility. Charging columns give mean stop counts; Sync. and Unsync. indicate stops overlapping or not with mandatory breaks or rests. Station-spacing rows use base-case EV settings (1.49 kWh/km, 750 kWh, 350 kW).

Figure 8 isolates the role of charging-station spacing. The results show that infrastructure density affects route-time competitiveness before it affects feasibility. When the spacing increases from 60 km to 140 km, the mean full-recharge penalty relative to the diesel proxy rises from 1.59 % to 3.46 %, while all tested scenarios remain feasible. Between approximately 140 km and 240 km, the penalty remains in the range of roughly 3.2 % to 4.0 %, suggesting that the routes can still be completed but charging is increasingly difficult to align with favorable break and rest opportunities. Beyond this range, feasibility begins to deteriorate: the feasibility rate falls to 97 % at 260 km and to 84.8 % at 300 km (Table 9), where the mean penalty among

feasible scenarios reaches 4.95 %. It should be noted that the small non-monotonic changes arise because station locations are regenerated for each spacing value, so larger spacings do not uniformly worsen all routes, and because the largest-spacing averages exclude infeasible cases.

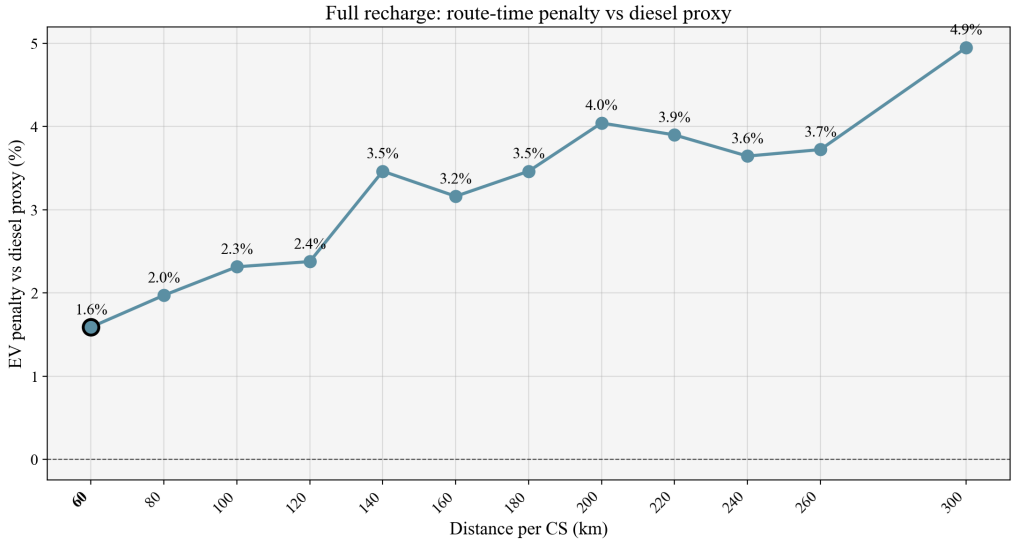


Figure 8: Mean paired route-time penalty of optimized full-recharge electric schedules relative to the diesel proxy as charging-station spacing increases. The highlighted marker indicates the base-case spacing.

6. Conclusion

This paper studied the joint scheduling of charging, mandatory breaks, and daily rests for battery-electric trucks operating on fixed multi-day long-haul routes. We introduced the electric vehicle truck driver scheduling problem as a multi-day extension of the truck driver scheduling setting in which energy replenishment becomes a binding operational resource, and we developed an exact solution approach by extending the parameter-free labeling framework of Goel and Irnich with battery dynamics and charging decisions. The resulting SPPRC formulation captures the interaction between Hours-of-Service compliance and energy feasibility without time discretization, and supports both full- and partial-recharge strategies within a unified framework.

The computational results show that the central operational issue is not charging per se, but charging *in synchrony* with legally mandated interruptions. In the base-case benchmark, partial recharge systematically improves on full recharge, while both exact methods substantially outperform the greedy benchmark. The absolute value of partial recharge grows with route length: although the percentage gain over full recharge remains modest, the time saving rises from about 12 minutes on short routes to about 49 minutes on long routes. Mechanism-based comparisons show that these gains arise because partial recharge allocates charging more flexibly across stops, increases the share of charging synchronized with breaks and rests, and reduces idle and unsynchronized charging on the routes where such effects matter most. Near-diesel

performance is achieved precisely in those instances where charging can be fully absorbed into the Hours-of-Service structure.

These findings carry two broader implications. First, the feasibility and competitiveness of long-haul electric trucking depend critically on whether charging opportunities can be embedded within the driver's regulatory timetable rather than added as separate time-consuming stops. Second, the value of partial recharge is highest in technologically tight settings, namely when energy consumption is high, battery capacity is smaller, or charging power is lower. In such regimes, the scheduling problem becomes more constrained, and the ability to tailor charging amounts to downstream breaks and rests becomes materially more valuable. From a planning perspective, this suggests that operational synchronization should be treated as a first-order design principle alongside battery size and charger deployment.

Several extensions remain relevant for future research. The present study considers fixed route sequences, homogeneous charging technology, deterministic travel and energy consumption. Extending the framework to integrated routing and scheduling decisions, heterogeneous and potentially congested charging infrastructure, nonlinear charging functions, and stochastic travel-time or energy-consumption processes would further strengthen its practical applicability.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI tools in order to suggest alternative formulations for individual sentences. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix A. Full-recharge SPPRC variant and pass-by arcs

needs to be updated validates, after finishing the partial charging description in the main text

This appendix presents the full-recharge variant of the SPPRC framework. The full-recharge strategy is recovered from the partial-recharge model by restricting every charging action to restore the battery to full capacity. Under this restriction, the energy state can be represented by a scalar resource, and additional pass-by arcs are used to preserve feasible continuations when a charging station is traversed without stopping.

Appendix A.1. Network structure with pass-by arcs

Figure A.9 illustrates the extended subnetwork for a route segment from customer n to customer m with a single intermediate charging station s. The corresponding arcs are (n, s) and (s, m).

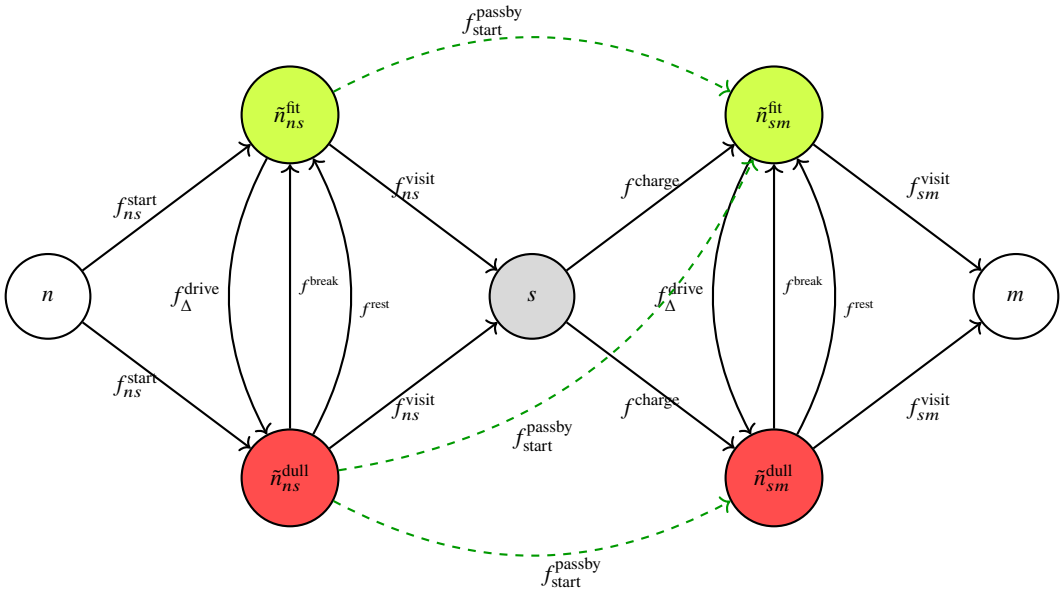


Figure A.9: Extended subnetwork with an intermediate charging station s and pass-by arcs under the full-recharge strategy.

At a charging-station node, the schedule may either initiate a charging activity or continue without charging. Charging is represented through charging arcs originating from the charging-station node, while pass-by arcs allow the station to be traversed without stopping. Together, these arcs preserve the feasible continuation of both fit and dull driver states while introducing the operational choice of whether and where to recharge.

The pass-by arcs determine how HoS states are propagated when a charging station is traversed without stopping. The fit-to-fit pass-by arc represents a driver who reaches the charging-station location in a state that permits further driving, either because no interruption is due or because the required break or rest has already been completed. If the station is skipped, the driver therefore remains fit and can immediately begin the arc from the charging-station location to m. A fit-to-dull pass-by arc is unnecessary, since it would require another break or rest before any additional driving; the dull state is reached naturally only when a

driving limit becomes binding on the subsequent arc. The dull-to-dull pass-by arc preserves an outstanding interruption requirement: after skipping the charging station, the driver remains dull and must take a break or rest before driving toward m. The dull-to-fit pass-by arc is a network connector that preserves the alternative continuations available after the charging-station node, allowing the next arc to begin with driving and, if necessary, a break or rest to be taken later. It does not reset any HoS resource, so such a continuation is retained only when it is feasible under the current driving and working-time counters.

Appendix A.2. Full-recharge label definition

Under the full-recharge strategy, each charging stop restores the battery to full capacity. The energy state can therefore be represented by a single scalar resource l_{reqEC}, which measures the energy requirement accumulated since the last recharge. Equivalently, the state of charge after any partial path is uniquely determined by $B - l_{\text{reqEC}}$, where B denotes the battery capacity.

A full-recharge label is thus defined as

$$\begin{aligned} l = (&l_{\text{time}},\ l_{\text{timeToNext}},\ l_{\text{drive|R}},\\ &l_{\text{drive|B}},\ l_{\text{elapsed|R}},\ l_{\text{elapsed|B}},\\ &l_{\text{reqEC}}) \end{aligned}$$

The resource l_{reqEC} is bounded by $0 \le l_{\text{reqEC}} \le B$. Its design is inspired by Desaulniers et al. (2016), but we retain an energy-based rather than time-based representation in order to preserve flexibility across different battery capacities, charging powers, and energy-consumption rates.

Appendix A.3. Full-recharge resource extension functions

For the full-recharge strategy, charging at a station cs is represented by an additional REF $f_{\text{cs}}^{\text{charge}}$. This extension resets the accumulated energy requirement l_{reqEC} to zero and increases the elapsed time by the charging duration implied by the current energy deficit and the available charging power. When charging overlaps with a legally admissible break or rest, the corresponding HoS counters are updated consistently so that synchronization between charging and regulatory interruptions is captured directly in the label propagation. In addition, pass-by arcs are modeled through $f_{\text{start}}^{\text{passby}}$, which allows a charging station to be traversed without stopping while preserving all resource values except the remaining travel time to the next route node. Table A.10 summarizes the resource updates for the REFs together with the full-recharge charging and pass-by extensions.

Appendix A.4. Full-recharge feasibility and dominance

Under the full-recharge strategy, energy feasibility is captured through the accumulated energy requirement since the last recharge. The scalar resource l_{reqEC} must remain within $[0, B]$ and is reset to zero by a recharge extension.

Table A.10: Resource updates for the REFs and the full-recharge charging extension.

Resource	f_{nm}^{start}	$f_{\Delta}^{\text{drive}}$	$f_{\Delta}^{\text{break}}$	f_{Δ}^{rest}	f_{nm}^{visit}	$f_{\text{cs}}^{\text{charge}}$		$f_{\text{start}}^{\text{passby}}$
l'_{time}	—	$l_{\text{time}} + \Delta$	$l_{\text{time}} + t_{\text{break}}$	$l_{\text{time}} + t_{\text{rest}}$	$l_{\text{time}} + s_m$	$\begin{cases} l_{\text{time}} \\ l_{\text{time}} + \max\left\{0, \frac{l_{\text{reqEC}}}{P^{\text{ch}}} - t_{\text{break}}\right\} \\ l_{\text{time}} + \frac{l_{\text{reqEC}}}{P^{\text{ch}}} \end{cases}$	$\begin{array}{l} l_{\text{elapsed}\|\text{R}} = 0 \\ l_{\text{elapsed}\|\text{B}} = 0 \\ \text{otherwise} \end{array}$	—
$l'_{\text{timeToNext}}$	t_{nm}	$l_{\text{timeToNext}} - \Delta$	—	—	—	$t_{\text{cs},m}$		$t_{\text{cs},m}$
$l'_{\text{drive}\|\text{R}}$	—	$l_{\text{drive}\|\text{R}} + \Delta$	—	0	—	—		—
$l'_{\text{drive}\|\text{B}}$	—	$l_{\text{drive}\|\text{B}} + \Delta$	0	0	—	—		—
$l'_{\text{elapsed}\|\text{R}}$	—	$l_{\text{elapsed}\|\text{R}} + \Delta$	$l_{\text{elapsed}\|\text{R}} + t_{\text{break}}$	0	$l_{\text{elapsed}\|\text{R}} + s_m$	$\begin{cases} 0 \\ l_{\text{elapsed}\|\text{R}} + \frac{l_{\text{reqEC}}}{P^{\text{ch}}} \end{cases}$	$\begin{array}{l} l_{\text{elapsed}\|\text{R}} = 0 \\ \text{otherwise} \end{array}$	—
$l'_{\text{elapsed}\|\text{B}}$	—	$l_{\text{elapsed}\|\text{B}} + \Delta$	0	0	$l_{\text{elapsed}\|\text{B}} + s_m$	$\begin{cases} 0 \\ l_{\text{elapsed}\|\text{B}} + \frac{l_{\text{reqEC}}}{P^{\text{ch}}} \end{cases}$	$\begin{array}{l} l_{\text{elapsed}\|\text{B}} = 0 \\ \text{otherwise} \end{array}$	—
l'_{reqEC}	—	$l_{\text{reqEC}} + \Delta \cdot \alpha$	—	—	—	0		—

Let l_1 and l_2 be two labels ending at the same node. Under the full-recharge strategy, l_1 dominates l_2 if

$$
\begin{aligned}
&l^{(1)}_{\text{time}} \le l^{(2)}_{\text{time}}, \qquad && l^{(1)}_{\text{timeToNext}} \le l^{(2)}_{\text{timeToNext}}, \\
&\mathbf{l}^{(1)}_{\text{drive}} \le \mathbf{l}^{(2)}_{\text{drive}}, \qquad && \mathbf{l}^{(1)}_{\text{elapsed}} \le \mathbf{l}^{(2)}_{\text{elapsed}}, \\
&l^{(1)}_{\text{reqEC}} \le l^{(2)}_{\text{reqEC}}.
\end{aligned}
$$

where

$$
\mathbf{l}_{\text{drive}} = \left(l_{\text{drive}|\text{R}},\, l_{\text{drive}|\text{B}}\right), \qquad \mathbf{l}_{\text{elapsed}} = \left(l_{\text{elapsed}|\text{R}},\, l_{\text{elapsed}|\text{B}}\right).
$$

If these componentwise conditions hold, l_2 can be discarded because it cannot yield a better feasible completion than l_1.

The scalar full-recharge dominance rule is therefore a special case of the partial-recharge dominance rule in the main text, obtained when the time and energy intervals collapse to singletons.

Algorithm 2 Benchmark instance generation

Require: Route class r, number of clusters K, number of customers N, allocation type a, random seed
Ensure: Instance data, distance matrix, time matrix, and diagnostics

Sample the total route length D according to route class r
Place K cluster centers approximately evenly along $[0, D]$, with small random perturbations
if $K = 1$ **then**
    Assign all customers to the single cluster
else if $a =$ BALANCED **then**
    Distribute the N customers approximately evenly among the clusters
else
    Select one dominant cluster at random
    Assign most customers to the dominant cluster and only a few to each remaining cluster
end if
Sample customer positions around their assigned cluster centers
Adjust customer positions to satisfy the required spacing constraints
Construct the predetermined visit sequence from the origin through the customers to the destination
Compute pairwise route distances d_{ij}
Compute travel times as $t_{ij} \leftarrow d_{ij}/60$, assuming a constant speed of 60 km/h
Generate customer service times
if the feasibility conditions are violated **then**
    Resample the instance until a feasible configuration is obtained
end if
Write the instance file, distance and time matrices, and diagnostic data

Appendix B. Pseudocode benchmark instance generator

Appendix C. Summary of benchmark instances

We should make this openly available on say Github. Add a link

Table C.11: Key columns only. Distances in km; times in h. Abbreviations: Test = scenario/test name; Class = route class (short/med/long); K = number of clusters; N = total customers; Var = variant (Balanced/Unbalanced); Dist_km = total route distance; Time_h = total route time; Cust/Cl = customers per cluster; AvgCl_km = average cluster distance; AvgIntra_km = average intra-cluster customer distance; AvgInter_km = average inter-cluster distance; AvgSvc_h = average service time per customer.

Test	Class	K	N	Var	Dist_km	Time_h	Cust/Cl	AvgCl_km	AvgIntra_km	AvgInter_km	AvgSvc_h
long_cl1_cu4	long	1	4	-	3929.38	56.1340	[4]	0.00	30.00	0.00	0.2650
long_cl2_cu2_Balanced	long	2	2	Balanced	4485.46	64.0780	[1,1]	674.75	0.00	674.75	0.4579
long_cl2_cu2_Unbalanced	long	2	2	Unbalanced	4472.53	63.8932	[1,1]	972.92	0.00	972.92	0.4207
long_cl2_cu3_Balanced	long	2	3	Balanced	4400.36	62.8623	[1,2]	797.97	37.73	797.97	0.4761
long_cl2_cu4_Balanced	long	2	4	Balanced	4162.88	59.4697	[2,2]	459.89	52.93	459.89	0.2409
long_cl2_cu4_Unbalanced	long	2	4	Unbalanced	4576.22	65.3745	[3,1]	1030.26	103.29	1030.26	0.2304
long_cl2_cu5_Balanced	long	2	5	Balanced	5119.54	73.1362	[3,2]	988.26	73.77	988.26	0.2437
long_cl3_cu12_Balanced	long	3	12	Balanced	5097.76	72.8252	[4,4,4]	810.85	45.65	1081.13	0.1286
long_cl3_cu12_Unbalanced	long	3	12	Unbalanced	4355.65	62.2236	[1,10,1]	647.62	30.00	863.49	0.1340
long_cl3_cu14_Balanced	long	3	14	Balanced	5063.93	72.3419	[5,5,4]	807.74	30.00	1076.99	0.1190
long_cl3_cu3_Balanced	long	3	3	Balanced	5016.56	71.6651	[1,1,1]	818.32	0.00	1091.09	0.3899
long_cl3_cu4_Balanced	long	3	4	Balanced	4155.45	59.3636	[2,1,1]	680.25	62.77	907.00	0.2582
long_cl3_cu4_Unbalanced	long	3	4	Unbalanced	5020.14	71.7163	[1,2,1]	758.57	50.84	1011.43	0.2508
long_cl3_cu5_Balanced	long	3	5	Balanced	5518.38	78.8340	[2,2,1]	894.65	44.15	1192.87	0.2550
med_cl1_cu2	med	1	2	-	2710.05	38.7150	[2]	0.00	84.81	0.00	0.4260
med_cl1_cu4	med	1	4	-	2615.02	37.3575	[4]	0.00	30.00	0.00	0.2420
med_cl2_cu2_Balanced	med	2	2	Balanced	2737.17	39.1024	[1,1]	446.85	0.00	446.85	0.4159
med_cl2_cu2_Unbalanced	med	2	2	Unbalanced	2321.67	33.1667	[1,1]	258.77	0.00	258.77	0.4043
med_cl2_cu3_Balanced	med	2	3	Balanced	3074.20	43.9171	[1,2]	499.00	43.26	499.00	0.4393
med_cl2_cu4_Balanced	med	2	4	Balanced	2413.01	34.4716	[2,2]	436.35	42.90	436.35	0.2751
med_cl2_cu4_Unbalanced	med	2	4	Unbalanced	3105.84	44.3691	[2,2]	681.02	73.82	681.02	0.2348
med_cl2_cu5_Balanced	med	2	5	Balanced	2251.60	32.1658	[3,2]	286.03	47.68	286.03	0.2607
med_cl3_cu11_Balanced	med	3	11	Balanced	2534.30	36.2043	[4,4,3]	332.33	56.27	443.11	0.1281
med_cl3_cu13_Balanced	med	3	13	Balanced	2513.67	35.9096	[5,4,4]	369.80	30.00	493.07	0.1271
med_cl3_cu13_Unbalanced	med	3	13	Unbalanced	3189.23	45.5604	[2,2,9]	437.50	36.03	583.33	0.1135
med_cl3_cu3_Balanced	med	3	3	Balanced	3292.08	47.0297	[1,1,1]	477.29	0.00	636.38	0.4144
med_cl3_cu4_Balanced	med	3	4	Balanced	2265.07	32.3581	[2,1,1]	360.47	142.64	480.63	0.2684
med_cl3_cu4_Unbalanced	med	3	4	Unbalanced	2200.53	31.4361	[2,1,1]	399.06	132.66	532.08	0.2710
med_cl3_cu5_Balanced	med	3	5	Balanced	3472.59	49.6084	[2,2,1]	559.40	34.92	745.87	0.2871
short_cl1_cu2	short	1	2	-	1081.84	15.4549	[2]	0.00	39.79	0.00	0.4019
short_cl2_cu2_Balanced	short	2	2	Balanced	1306.92	18.6703	[1,1]	213.84	0.00	213.84	0.4766
short_cl2_cu2_Unbalanced	short	2	2	Unbalanced	1512.53	21.6076	[1,1]	221.86	0.00	221.86	0.4272
short_cl2_cu3_Balanced	short	2	3	Balanced	1109.01	15.8429	[1,2]	108.55	72.96	108.55	0.3558
short_cl2_cu4_Balanced	short	2	4	Balanced	1221.36	17.4480	[2,2]	162.51	84.56	162.51	0.2685
short_cl2_cu4_Unbalanced	short	2	4	Unbalanced	1408.63	20.1233	[1,3]	159.19	93.35	159.19	0.2364
short_cl2_cu5_Balanced	short	2	5	Balanced	1249.49	17.8499	[3,2]	175.63	35.51	175.63	0.2610
short_cl3_cu10_Balanced	short	3	10	Balanced	1761.42	25.1631	[4,3,3]	260.70	30.00	347.60	0.1333
short_cl3_cu12_Balanced	short	3	12	Balanced	1941.81	27.7401	[4,4,4]	277.09	43.40	369.45	0.1206
short_cl3_cu12_Unbalanced	short	3	12	Unbalanced	1541.88	22.0269	[8,2,2]	215.62	43.30	287.50	0.1268
short_cl3_cu3_Balanced	short	3	3	Balanced	1450.69	20.7242	[1,1,1]	234.78	0.00	313.04	0.4172
short_cl3_cu4_Balanced	short	3	4	Balanced	1270.79	18.1542	[2,1,1]	241.75	106.66	322.33	0.2682
short_cl3_cu4_Unbalanced	short	3	4	Unbalanced	1324.85	18.9264	[2,1,1]	197.50	31.86	263.33	0.2117

short_cl3_cu5_Balanced	short	3	5	Balanced	1876.99	26.8142	[2,2,1]	259.19	68.40	345.58	0.2621

Appendix D. Additional Per-Instance Comparison

Table D.12: Near-diesel diagnostic cases in the base-case setting.

Instance	EV (h)	Diesel (h)	CS	Sync. (%)	Unsync.	Idle (min)
s_cl2_cu4_U	27.86	27.86	3	100.0	0	0
s_cl2_cu3_B	27.19	27.19	2	100.0	0	0
m_cl2_cu2_U	68.95	68.95	6	100.0	0	0
m_cl3_cu4_U	64.10	64.10	5	100.0	0	0
s_cl3_cu10_B	44.20	44.20	4	100.0	0	0
s_cl2_cu4_B	44.19	44.19	2	100.0	0	0

These are the base-case instances for which the optimized electric schedule attains the same total route time as the diesel proxy. In all six cases, every charging stop is synchronized with a mandatory break or rest.

Table D.13: Full- and partial-recharge results from the computational experiments. Times are in hours. Abbreviations: ID = short instance identifier used in the results files; Test = full instance name; F_Time = full-recharge route time; F_CS = number of charging stops under full recharge; F_Sync = number of synchronized charging stops under full recharge; F_Unsync = number of unsynchronized charging stops under full recharge; F_Break = number of full-recharge charging stops synchronized with breaks; F_Rest = number of full-recharge charging stops synchronized with daily rests; P_Time = partial-recharge route time; P_CS = number of charging stops under partial recharge; P_Sync = number of synchronized charging stops under partial recharge; P_Unsync = number of unsynchronized charging stops under partial recharge; P_Break = number of partial-recharge charging stops synchronized with breaks; P_Rest = number of partial-recharge charging stops synchronized with daily rests.

ID	Test	F_Time	F_CS	F_Sync	F_Unsync	F_Break	F_Rest	P_Time	P_CS	P_Sync	P_Unsync	P_Break	P_Rest
S_cl1_cu2_B	short_cl1_cu2	31.73	3	1	2	0	1	31.49	3	1	2	0	1
S_cl2_cu3_B	short_cl2_cu3_Balanced	27.27	1	1	0	0	1	27.19	2	2	0	1	1
S_cl2_cu4_B	short_cl2_cu4_Balanced	44.35	2	2	0	0	2	44.19	2	2	0	0	2
S_cl2_cu4_U	short_cl2_cu4_Unbalanced	27.94	1	1	0	0	1	27.86	3	3	0	2	1
S_cl2_cu5_B	short_cl2_cu5_Balanced	30.88	2	1	1	0	1	30.72	2	1	1	0	1
S_cl3_cu10_B	short_cl3_cu10_Balanced	44.36	2	2	0	0	2	44.20	4	4	0	2	2
S_cl3_cu12_B	short_cl3_cu12_Balanced	43.69	2	2	0	0	2	43.53	4	4	0	2	2
S_cl3_cu12_U	short_cl3_cu12_Unbalanced	30.57	2	1	1	0	1	30.16	2	2	0	1	1
S_cl3_cu3_B	short_cl3_cu3_Balanced	31.35	3	1	2	0	1	30.86	3	2	1	1	1
S_cl3_cu4_B	short_cl3_cu4_Balanced	31.18	2	1	1	0	1	31.02	3	2	1	1	1
S_cl3_cu4_U	short_cl3_cu4_Unbalanced	32.17	2	1	1	0	1	31.85	3	2	1	1	1
S_cl3_cu5_B	short_cl3_cu5_Balanced	31.48	2	1	1	0	1	31.32	3	2	1	1	1
M_cl1_cu2_B	med_cl1_cu2	68.95	6	6	0	0	6	68.95	6	6	0	0	6
M_cl1_cu4_B	med_cl1_cu4	64.26	5	4	1	0	4	64.10	5	5	0	1	4
M_cl2_cu2_B	med_cl2_cu2_Balanced	69.44	4	3	1	0	3	68.95	6	6	0	0	6
M_cl2_cu2_U	med_cl2_cu2_Unbalanced	37.11	2	2	0	0	2	36.62	3	3	0	1	2
M_cl2_cu3_B	med_cl2_cu3_Balanced	47.95	3	3	0	0	3	47.46	4	4	0	1	3
M_cl2_cu4_B	med_cl2_cu4_Balanced	64.09	5	5	0	0	5	63.60	7	7	0	2	5
M_cl2_cu4_U	med_cl2_cu4_Unbalanced	65.70	4	3	1	0	3	65.20	6	5	1	2	3
M_cl2_cu5_B	med_cl2_cu5_Balanced	37.37	2	2	0	0	2	36.88	3	3	0	1	2
M_cl3_cu11_B	med_cl3_cu11_Balanced	47.45	2	2	0	0	2	47.29	4	4	0	2	2
M_cl3_cu13_B	med_cl3_cu13_Balanced	48.39	2	2	0	0	2	48.23	4	4	0	2	2
M_cl3_cu13_U	med_cl3_cu13_Unbalanced	48.98	2	2	0	0	2	48.82	4	4	0	2	2
M_cl3_cu3_B	med_cl3_cu3_Balanced	48.66	4	3	1	0	3	47.84	5	4	1	1	3
M_cl3_cu4_B	med_cl3_cu4_Balanced	38.37	2	2	0	0	2	37.88	3	3	0	1	2

M_cl3_cu4_U	med_cl3_cu4_Unbalanced	33.00	2	2	0	0	2	32.51	3	3	0	1	2
M_cl3_cu5_B	med_cl3_cu5_Balanced	50.26	3	2	1	0	2	49.77	4	3	1	1	2
L_cl1_cu2_B	long_cl1_cu2	74.61	6	4	2	0	4	74.12	7	5	2	1	4
L_cl1_cu4_B	long_cl1_cu4	86.24	5	3	2	0	3	85.75	7	5	2	2	3
L_cl2_cu2_B	long_cl2_cu2_Balanced	75.95	4	3	1	0	3	74.65	7	6	1	2	4
L_cl2_cu2_U	long_cl2_cu2_Unbalanced	76.55	4	3	1	0	3	75.06	6	5	1	2	3
L_cl2_cu3_B	long_cl2_cu3_Balanced	76.02	5	3	2	0	3	74.53	7	5	2	2	3
L_cl2_cu4_B	long_cl2_cu4_Balanced	65.36	4	3	1	0	3	64.05	5	4	1	1	3
L_cl2_cu4_U	long_cl2_cu4_Unbalanced	79.11	6	3	3	0	3	77.81	8	5	3	2	3
L_cl2_cu5_B	long_cl2_cu5_Balanced	88.17	5	3	2	0	3	87.68	7	5	2	2	3
L_cl3_cu12_B	long_cl3_cu12_Balanced	74.79	4	4	0	0	4	74.63	8	8	0	4	4
L_cl3_cu12_U	long_cl3_cu12_Unbalanced	87.07	5	3	2	0	3	86.58	6	4	2	1	3
L_cl3_cu14_B	long_cl3_cu14_Balanced	73.33	4	4	0	0	4	73.16	8	8	0	4	4
L_cl3_cu3_B	long_cl3_cu3_Balanced	88.04	6	4	2	0	4	86.75	7	5	2	1	4
L_cl3_cu4_B	long_cl3_cu4_Balanced	76.94	5	3	2	0	3	75.45	7	5	2	2	3
L_cl3_cu4_U	long_cl3_cu4_Unbalanced	85.90	7	4	3	0	4	84.41	9	6	3	2	4
L_cl3_cu5_B	long_cl3_cu5_Balanced	98.45	6	4	2	0	4	97.96	8	6	2	2	4